A CONTINUOUS ALMOST COCONTINUOUS FUNCTOR PRESERVES FINITE CONNECTED COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Colimits of sheaves are sheafifications of colimits of presheaves. Same chapter, Lemma lemma-colimit-sheaves. Identifies the colimit of a diagram of sheaves with the sheafification of the presheaf colimit.
- Sheafification in the presence of almost cocontinuity. Same chapter, Lemma lemma-almost-cocontinuous-sheafification. Supplies the commutation of the direct image with sheafification for presheaves satisfying the relevant condition.

Source reference. Original lemma 00WI: sources/stacks-sites/pinned-original-source.tex, source lines 10262–10262. Navigation pointer invoked at proof line 10262; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04B8: sources/stacks-sites/pinned-original-source.tex, source lines 10267–10267. Navigation pointer invoked at proof line 10267; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$, $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that u is continuous and almost cocontinuous. Then $u^s = u^p : Sh(\mathcal{D}) \to Sh(\mathcal{C})$ commutes with pushouts and coequalizers (and more generally finite connected colimits).*

Proof. Let $\mathcal{I}$ be a finite connected index category. Let $\mathcal{I} \to Sh(\mathcal{D})$, $i \mapsto \mathcal{G}_i$ by a diagram. We know that the colimit of this diagram is the sheafification of the colimit in the category of presheaves, see Lemma 00WI. Denote colim^{Psh} the colimit in the category of presheaves. Since $\mathcal{I}$ is finite and connected we see that $\text{colim}_i^{Psh} \mathcal{G}_i$ is a presheaf satisfying the assumptions of Lemma 04B8 (because a finite connected colimit of singleton sets is a singleton). Hence that lemma gives

$$\begin{aligned} u^s(\text{colim}_i \mathcal{G}_i) &= u^s((\text{colim}_i^{Psh} \mathcal{G}_i)^\#) \\ &= (u^p(\text{colim}_i^{Psh} \mathcal{G}_i))^\# \\ &= (\text{colim}_i^{PSh} u^p(\mathcal{G}_i))^\# \\ &= \text{colim}_i u^s(\mathcal{G}_i) \end{aligned}$$

as desired. □